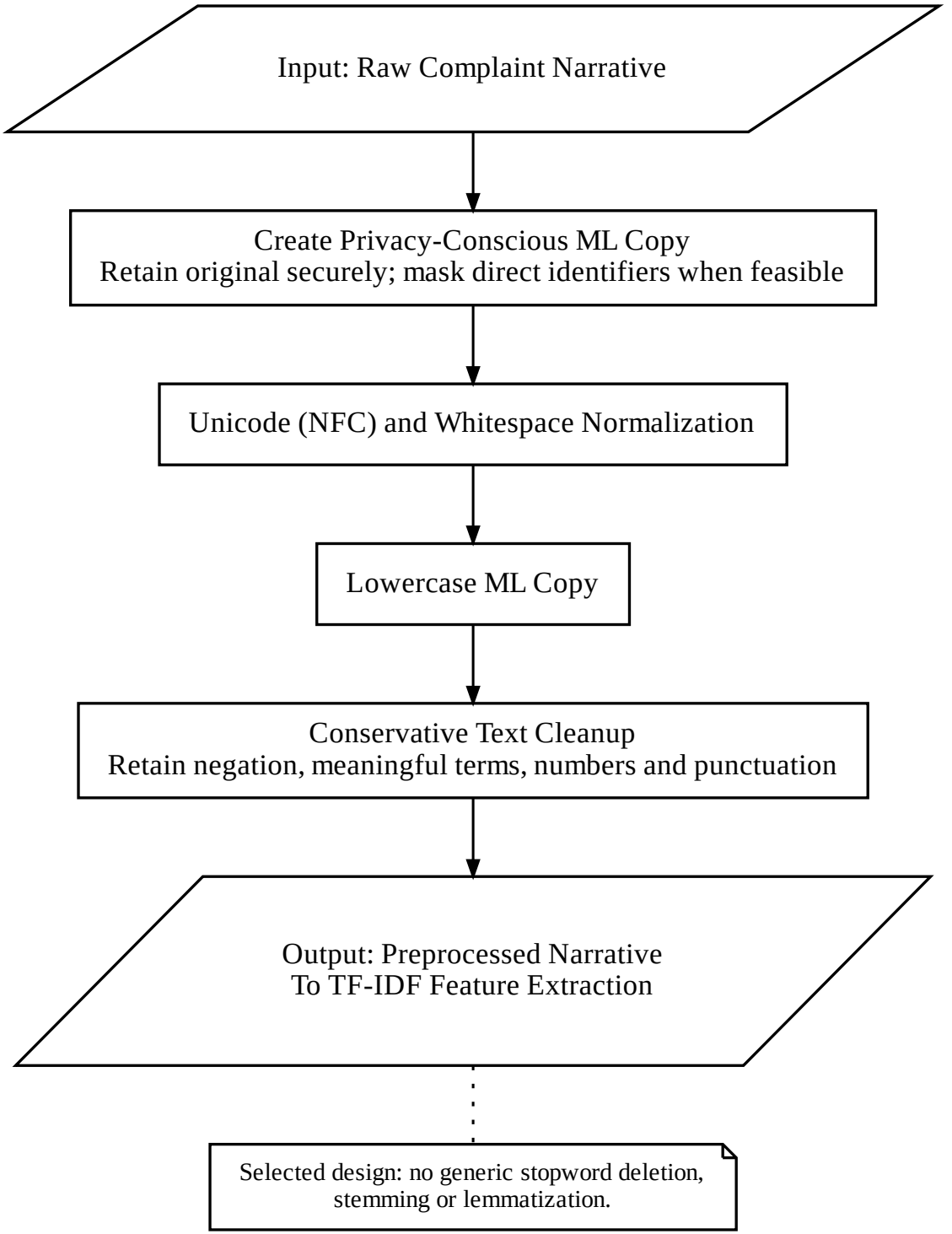
Input: Raw Complaint Narrative
Create Privacy-Conscious ML Copy
Retain original securely; mask direct identifiers when feasible
Unicode (NFC) and Whitespace Normalization
Lowercase ML Copy
Conservative Text Cleanup
Retain negation, meaningful terms, numbers and punctuation
Output: Preprocessed Narrative
To TF-IDF Feature Extraction
Selected design: no generic stopword deletion,
stemming or lemmatization.